Kidney Stone Ontology: From Clinical Notes to Structured Registry

A walkthrough of the data model, its design rationale, and how raw text becomes answerable research data

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Definitions

Reference page. Every term in this document, one line each.

Clinical Event — One clinical fact extracted from a note. A stone on imaging, a drug given, a procedure done. The central object in this system.

Family — One of 5 clinical domains that group related event types: history_timeline, symptoms_imaging, procedure_devices, medications, outcomes_complications.

Event Type — One of 10 specific kinds of clinical facts within a family. Each family contains 1–3 event types. The event type determines which attributes that event carries.

Preferred Term — The canonical name for what an event describes. “ureteral stone”, “ureteroscopy”, “tamsulosin”. Normalized through Event Terms.

Concept ID — A stable identifier built from family + event_type + preferred_term. Example: `symptoms_imaging.imaging_finding.ureteral_stone`

Attributes — Key-value pairs that describe the details of an event. laterality=RIGHT, size=5, unit=mm. Each event type has its own attribute set.

Evidence Set — The exact quoted text from the note with character offsets and section name. Every event traces back to the sentence it came from.

Assertion Strength — How certain the clinician was. CONFIRMED = definitely present. POSSIBLE = suspected. NEGATED = explicitly ruled out.

Temporal Status — When relative to this visit. CURRENT = happening now. HISTORICAL = happened before. FUTURE = planned.

Experiencer — Who this happened to. PATIENT or FAMILY_MEMBER (for family history).

Event Terms — 990 aliases that map to 252 canonical preferred terms. Normalizes the name of the thing. “UVJ calculus” → “ureteral stone”.

Attribute Terms — 910 aliases that map to 327 canonical attribute values. Normalizes the details. “right” → RIGHT, “ureterovesicular junction” → UVJ.

Normalization Status — CANONICAL = alias found in vocab. PROVISIONAL = new term, flagged for review. CUSTOM = has a user-provided concept_id.

Attribute Policy — 89 rules defining what values are allowed per attribute per event type. Each rule specifies kind, style, and provisional.

> **attribute_kind** — controlled = must be in vocab. semi_structured = follows a pattern (dates, measurements). free_text = any value allowed.
>
> **canonical_style** — How the value is formatted. upper_tokens = RIGHT, LEFT. lower_text = nonobstructing. clinical_mixed = UVJ, UPJ. Others: measurement_unit, numeric_or_text, date_or_span, free_text.
>
> **allow_provisional** — true = unrecognized values accepted and flagged. false = unrecognized values rejected outright.

Relation — A typed link between two events in the same note. 9 types: performed_for, caused_by, follow_up_for, treated_by, decompresses, temporally_linked, modifies, resolved_after, causes_obstruction.

Projection — The process of computing answers from events. Events go in, answers to clinician questions come out.

Resolver — The function that reads events and extracts the right data to answer one question. Each of the 130 questions has exactly one resolver.

- **event_presence** — Checks if a matching event exists. Returns yes/no. Example: “Was a stent placed?”
- **event_attribute_values** — Reads a specific attribute from matching events. Example: “Where is the stone?” → reads imaging_finding.location.
- **event_concepts** — Matches events against keyword lists. Example: “Did symptoms resolve?” → matches outcome_event against [“symptom resolution”, “pain resolved”].
- **event_grouped_counts** — Counts events grouped by a field. Example: “How many medications?” → counts medication_exposure events.

Registry Question — One of 130 clinician-designed questions. Each has a registry_id, a resolver, and a priority score from clinician voting.

- **registry_id** — Unique identifier for each question. Example: `stone_location`, `was_stent_placed`.
- **Resolution State** — Whether a question was answered or unanswered for a given note. Unanswered means the note had no events matching the resolver.
- **Answer Scope** — What data the question looks at. note = this note only. patient_timeline = all notes for this patient. encounter = this visit. *(expanded)*

Ontology Taxonomy Tree

The full schema. Five families, ten event types, and all 252 canonical preferred terms defined in the ontology. Attributes show their policy type: controlled, semi_structured, or free_text.

Kidney Stone Ontology — 5 families · 10 event types · 252 preferred terms · 89 attribute policies · 9 relation types

history_timeline

Prior history, family history, longitudinal stone history, social/support context

history_fact (15 preferred terms · 7 attributes)

chronic kidney disease | family history of kidney stones | horseshoe kidney | kidney stone history | kidney stones during pregnancy | medullary sponge kidney | multidrug-resistant organism UTI | obstetric complication | prior lithotripsy | prior nephrostomy | prior percutaneous nephrolithotomy | prior ureteral stent | prior ureteroscopy | recurrent urinary tract infection | solitary kidney

status | count | timeframe | history_kind | subject | age_of_onset | intervention_status

social_context_fact (14 preferred terms · 3 attributes)

caregiver support | disability | education level | employed | employment status | housing instability | limited support | lives alone | no support | social isolation | transportation available | transportation barrier | unemployed | uninsured

support_type | value | timeframe

symptoms_imaging

Stone-related symptoms, imaging findings, stone features

symptom_presentation (1 preferred term · 6 attributes)

stone-related symptomatic presentation

symptoms | laterality | severity | clinical_context | exam_finding | onset_duration

imaging_finding (23 preferred terms · 15 attributes · 3 relations)

bladder stone | calcium oxalate stone | calcium phosphate stone | calyceal diverticulum | calyceal rupture | cystine stone | delayed nephrogram | forniceal rupture | hydronephrosis | hydroureter | hydroureteronephrosis | kidney stone | medullary nephrocalcinosis | obstructive uropathy | perinephric stranding | periureteral stranding | renal parenchymal thinning | staghorn calculus | struvite stone | ureteral stone | ureteral stricture | uric acid stone | urinoma

laterality | location | severity | size | unit | hounsfield_unit | attribution | status | anatomic_detail | calyceal_detail | stone_count | stone_burden | comparison_status | comparison_reference | clinical_significance

procedure_devices

Procedures, devices, stents, operative details

procedure_event (23 preferred terms · 21 attributes · 4 relations)

cystoscopy | electrohydraulic lithotripsy | extracorporeal shock wave lithotripsy | laser lithotripsy | lithotripsy | nephrectomy | nephrostogram | nephrostomy exchange | nephrostomy internalization | nephrostomy placement | nephrostomy removal | nephroureterectomy | open nephrolithotomy | percutaneous access | percutaneous nephrolithotomy | pyeloplasty | retrograde pyelogram | stone basket retrieval | ultrasonic lithotripsy | ureteral stent exchange | ureteral stent placement | ureteral stent removal | ureteroscopy

laterality | status | approach | procedure_date | qualifier | indication | target_finding | anatomic_target | laser_power | laser_frequency | laser_type | laser_fiber_type | laser_fiber_size | dilation_diameter | irrigation_pressure | suction_pressure | shock_rate | max_energy | total_energy | swl_machine_type | lithotripter_energy_source

device_event (20 preferred terms · 11 attributes · 5 relations)

access sheath | catheter | cystoscope | dual lumen catheter | foley catheter | grasping forceps | guidewire | laser fiber | nephroscope | nephrostomy tube | nephroureteral stent | percutaneous access needle | pigtail catheter | stone retrieval basket | suprapubic catheter | surgical drain | ureteral access sheath | ureteral catheter | ureteral stent | ureteroscope

device_type | laterality | status | size | length | unit | indication | target_finding | anatomic_target | diameter | make_model

medications

Medication administrations and prescriptions

medication_exposure (88 preferred terms · 9 attributes · 1 relation)

IV antibiotics | IV fluids | acetaminophen | acetaminophen-codeine | alfuzosin | allopurinol | amoxicillin | amoxicillin-pot clavulanate | ampicillin-sulbactam | azithromycin | bupivacaine | calcium citrate | cefazolin | cefepime | cefpodoxime | ceftriaxone | cefuroxime | celecoxib | cephalexin | chlorthalidone | ciprofloxacin | clindamycin | colchicine | d-penicillamine | dexamethasone | diclofenac | diphenhydramine | doxazosin | dutasteride | enoxaparin | ertapenem | febuxostat | fentanyl | finasteride | fosfomycin | gabapentin | gentamicin | granisetron | hydrochlorothiazide | hydrocodone | hydrocodone-acetaminophen | hydromorphone | ibuprofen | indomethacin | ketorolac | lactated ringers | levofloxacin | lidocaine | lisinopril | meloxicam | meropenem | methylprednisolone | metoclopramide | metronidazole | mirabegron | morphine | naproxen | nitrofurantoin | normal saline | ondansetron | oxybutynin | oxycodone | oxycodone-acetaminophen | phenazopyridine | piperacillin-tazobactam | potassium chloride | potassium citrate | prednisone | pregabalin | prochlorperazine | promethazine | silodosin | sodium bicarbonate | sodium citrate-citric acid | solifenacin | spironolactone | tamsulosin | terazosin | thiazide diuretic | tiopronin | tolterodine | tramadol | trimethoprim | trimethoprim-sulfamethoxazole | trospium | vancomycin | vitamin D | warfarin

drug_class | dose | route | frequency | duration | date | indication | context | status

outcomes_complications

Complications, outcomes, recommendations, follow-up

adverse_effect (29 preferred terms · 5 attributes · 2 relations)

acute kidney injury | allergic reaction | anaphylaxis | anesthetic complication | bacteremia | bladder spasm | bleeding | clot retention | contrast reaction | dizziness | fever | gross hematuria | hives | nausea | nausea and vomiting | obstructive nephropathy | perinephric hematoma | pyelonephritis | pyonephrosis | rash | sepsis | steinstrasse | subcapsular hematoma | ureteral injury | urinary retention | urinary tract infection | urosepsis | vomiting | wound complication

reaction_type | severity | suspected_cause | timing | status

outcome_event (16 preferred terms · 7 attributes · 3 relations)

culture negative | culture positive | fever resolution | infection resolved | no stone passage | pain controlled | partial stone clearance | readmission | renal function improved | renal function stable | residual stone | stent intolerance | stone free | stone passage | stone recurrence | symptom resolution

status | timing | value | qualifier | count | comparison_status | comparison_reference

recommendation_event (23 preferred terms · 5 attributes · 2 relations) ***(expanded)***

24-hour urine collection | activity modification | dietary modification | elective extracorporeal shock wave lithotripsy | elective percutaneous nephrolithotomy | elective ureteroscopy | hydration | low oxalate diet | low purine diet | low sodium diet | medical expulsive therapy | metabolic workup | nephrology referral | pharmacotherapy | primary care follow-up | repeat imaging | return precautions | stent removal | stone analysis | strain urine for stone | thiazide therapy | urology follow-up | urology referral

recommendation_type | status | target | qualifier | indication

Cross-cutting Relations (9 types linking events within the same note)

performed_for | caused_by | follow_up_for | treated_by | decompresses | temporally_linked | modifies | resolved_after | causes_obstruction

1. The Problem

A note states: "Punctate 2 mm nonobstructing calculus at the right ureterovesicular junction." The question is how do you take that and make it structured. How do you get from a sentence a surgeon dictated into a queryable fact with a laterality, a location, a size, and a canonical name that matches every other note that mentions the same kind of stone.

2. The Data Model: Why These Elements

The central object in this system is the **clinical event**. An event represents one clinical fact extracted from a note: a stone found on imaging, a drug administered, a procedure performed, a complication documented. The model was designed around one question: what are all the distinct things a clinician writes about in a kidney stone note that a researcher would want to query later?

Kidney stone care follows a natural clinical arc. Five families cover it. Each one maps to a concern urologists think about as a unit.

Family	Clinical Concern	Event Types	What It Captures
history_timeline	What happened before this visit?	history_fact, social_context_fact	Prior stones, family history, anatomic variants, caregiver/ transportation barriers
symptoms_imaging	What are we seeing now?	symptom_presentation, imaging_finding	Flank pain, hematuria, stone location/size/ laterality, hydronephrosis, obstruction
procedure_devices	What did we do?	procedure_event, device_event	Ureteroscopy, PCNL, stent placement, nephrostomy tubes, laser settings
medications	What drugs were given?	medication_exposure	Drug name, dose, route, frequency, indication, context (ED, home, intra-op)
outcomes_complications	What was the result?	adverse_effect, outcome_event, recommendation_event	Complications, stone-free status, symptom resolution, follow-up instructions

> Fewer families means less ambiguity about where a concept belongs. Symptoms and imaging go together because clinicians evaluate them together.

Each event type captures a distinct clinical fact. Ten types across five families. A family is the clinical domain, the event type is the specific thing inside that domain. procedure_devices has two event types because the surgery itself and the stent left behind are different facts with different attributes and different research questions. The table below maps every type to its parent family.

Event Type	Family	What It Represents	Example
history_fact	history_timeline	A prior clinical fact: stone history, prior surgery, family history	"history of nephrolithiasis"
social_context_fact	history_timeline	Social support, transportation, caregiver availability	"caregiver present"
symptom_presentation	symptoms_imaging	Stone-related symptoms the patient presented with	"acute right flank pain with nausea"
imaging_finding	symptoms_imaging	A finding on imaging: stone, hydronephrosis, obstruction	"5mm calculus at right UVJ"
procedure_event	procedure_devices	A surgery or interventional procedure performed	"ureteroscopy with laser lithotripsy"
device_event	procedure_devices	A device placed, removed, or currently in the patient	"6F ureteral stent placed"
medication_exposure	medications	One drug administration or prescription	"tamsulosin 0.4 mg PO daily"
adverse_effect	outcomes_complications	A complication or side effect of treatment	"acute kidney injury post-operatively"
outcome_event	outcomes_complications	A treatment outcome: stone passage, symptom resolution	"stone-free on follow-up imaging"
recommendation_event	outcomes_complications	A follow-up or lifestyle recommendation	"follow up in urology clinic"

What Goes Inside an Event

Every event carries the same core fields. These were chosen so that each event is self-contained: you can look at a single event and know what it is, when it happened relative to this visit, how certain the clinician was, and exactly where in the note it came from.

Field	What It Records	Example
event_id	Unique ID within this note	EV1, EV2, EV3...
family	Which of the 5 clinical domains	symptoms_imaging
event_type	Which of the 10 types	imaging_finding
preferred_term	Canonical name for the clinical concept	ureteral stone
attributes	Key-value pairs specific to the event type	{laterality: RIGHT, size: 5, unit: mm}
temporal_status	When relative to this visit	CURRENT, HISTORICAL, or FUTURE
assertion_strength	How certain the clinician was	CONFIRMED, POSSIBLE, or NEGATED
experiencer	Who this happened to	PATIENT or FAMILY_MEMBER
evidence_set	Exact quoted text with character offsets and section	"5mm calculus..." at chars 1204-1261
relations	Links to other events in the same note	performed_for -> EV1

"Kidney stone" means three different things depending on context. A CURRENT stone is why the patient is here today. A HISTORICAL stone means they are a recurrent stone former. A NEGATED stone ("no evidence of calculus") means imaging was performed and nothing was found. That is clinically different from never having been imaged. Without temporal_status and assertion_strength, a query cannot distinguish these cases.

How the Pieces Connect

Below is every step that turns a sentence into a queryable fact. One real sentence from an ED note, every transformation, real data at every tier.

Clinical Note is the raw input. A surgeon dictated "Punctate 2 mm nonobstructing calculus at the right ureterovesicular junction" and that sentence needs to become structured data. **Event Terms** normalize the name of the thing. The note says "UVJ calculus" but across 9,869 notes that same stone shows up as "ureterovesicular junction stone" and "distal ureteral calculus" so 990 aliases collapse to 252 canonical names and every spelling lands on one term. **Attribute Terms** normalize the details about that thing. "right" becomes RIGHT, "ureterovesicular junction" becomes UVJ. 910 aliases map to 327 canonical values. **Assembled Event** is what comes out. It has a family (the clinical domain like symptoms_imaging), an event_type (the specific thing inside that domain like imaging_finding), a preferred_term from Event Terms, and all the normalized attributes from Attribute Terms. Each family contains one to three event types and the event type determines which attributes that event carries. **Attribute Policy** is the rulebook. 89 rules say which values are allowed for each attribute on each event type. Take laterality on an imaging_finding: the policy says kind=controlled, style=upper_tokens, provisional=false. That means only LEFT, RIGHT, BILATERAL are legal. The model produces "right", normalization maps it to RIGHT, policy checks RIGHT against the vocab, it passes. If the model had produced "both sides" and normalization could not map it, provisional=false means reject it outright. That value never enters the dataset. **Relations** connect events within the same note. A medication was given for a stone, a stent was placed during a procedure. When one event is not enough to answer a question you follow the links. **Projection** is where events become answers. A resolver is just a function that reads events and pulls the right data out to answer a specific question. The clinician asks "Where is the stone?" The resolver for that question is event_attribute_values. It finds all imaging_finding events in the note, reads their location attribute, and returns UVJ. A different question like "Was a stent placed?" uses event_presence. It looks for device_event events with device_type=ureteral stent and returns yes or no. Four resolver types total: event_presence (does a thing exist), event_attribute_values (read a specific field), event_concepts (match events against keyword lists), event_grouped_counts (count across multiple events). **Registry Questions** are the 130 questions clinicians designed and prioritized. Each question has exactly one resolver wired to it and the resolver knows which event types to query and which attributes to read.

Clinical Note — free text input

```
Punctate 2 mm nonobstructing calculus at the right ureterovesicular junction.
```

↓ the model reads this and proposes a raw event

Event Terms — 990 aliases → 252 canonical

raw term:	~~"UVJ calculus"~~ → **"ureteral stone"**
status:	**CANONICAL** (alias found)
on miss:	PROVISIONAL, flagged for review

Attribute Terms — 910 aliases → 327 canonical

laterality:	~~"right"~~ → **RIGHT** upper_tokens
location:	~~"ureterovesicular jx"~~ → **UVJ** clinical_mixed
status:	~~"nonobstructing"~~ → **nonobstructing** lower_text
stone_burden:	~~"punctate"~~ → **punctate** lower_text
size / unit:	2 / mm numeric, kept as-is

left card normalizes the name of the thing, right card normalizes the details about it

Assembled Event

family → event_type → attributes

family → type: symptoms_imaging → imaging_finding (1–3 types per family)
preferred_term: **ureteral stone** NORMALIZED
concept_id: symptoms_imaging.imaging_finding.ureteral_stone GENERATED

assertion: CONFIRMED
temporal: CURRENT
attributes: {laterality: **RIGHT**, location: **UVJ**, status: **nonobstructing**, size: 2, unit: mm, stone_burden: **punctate**}
evidence: "Punctate 2 mm nonobstructing calculus..." chars 847–921, Radiology section

↓ now check if those values are actually allowed and link events that belong together

Attribute Policy

89 rules: kind + style + provisional

laterality: controlled, upper_tokens, provisional=false ✓ RIGHT in vocab
location: controlled, clinical_mixed, provisional=false ✓ UVJ in vocab
status: controlled, lower_text, provisional=false ✓ in vocab
size: free_text ✓ any value
example fail: "both sides" → unmappable, provisional=false → rejected

Relations

9 types linking events in this note

note events: 7 total (1 stone, 1 symptom, 4 meds, 1 rec)
EV3–EV6: medication → **indication_for** → EV1 (stone)
EV7: recommendation → **follow_up_for** → EV1
effect: query "meds for ureteral stones" follows links

↓ everything is normalized and validated, now answer the clinician's questions

Projection

answers traced to events

Q: "Where is the stone?"
resolver: event_attribute_values → imaging_finding.location
answer: **UVJ** backed by EV1

Q: "What is the laterality?"
resolver: event_attribute_values → imaging_finding.laterality
answer: **RIGHT** backed by EV1

Q: "Was a stent placed?"
resolver: event_presence → device_event where device_type=ureteral stent
answer: **yes** backed by EV2

exclusion: NEGATED or HISTORICAL events do not produce answers

Registry Questions

130 clinician-designed

this event: answers **2** of 9 symptoms_imaging questions
this note: 7 events answer **8** questions across 3 families

stored as: registry_id + answer + supporting event IDs
audit trail: answer → event → quoted text → section → offsets

each question has one resolver. the resolver finds matching events and reads the right fields to produce an answer.

3. What Each Event Type Captures

Each event type has its own set of attributes. These are the specific details a clinician documents and a researcher needs. The attributes were chosen by reviewing the 130 registry questions and asking: what structured fields do we need in order to answer each question? For example, the question "What is the laterality of the stone?" requires that imaging_finding events carry a laterality attribute.

Imaging Finding

The most attribute-rich event type, because stone characterization drives treatment decisions. A 2mm nonobstructing stone is managed differently from a 2cm staghorn calculus.

Attribute	What It Records	Example Values
laterality	Which side	LEFT, RIGHT, BILATERAL
location	Anatomic position	ureter, kidney, UVJ, UPJ, renal pelvis, bladder
size	Measurement	5 (paired with unit: mm)
status	Obstructing or not	obstructing, nonobstructing, partially obstructing
stone_burden	How many or how much	single, multiple, staghorn, punctate
severity	Grade if stated	mild, moderate, severe (for hydronephrosis)
hounsfield_unit	CT density	800 HU

Symptom Presentation

Captures the symptom cluster the patient presented with. One event per symptomatic episode, not one event per symptom, because clinicians describe them as a group.

Attribute	What It Records	Example Values
symptoms	List of presenting symptoms	flank pain, nausea, hematuria, dysuria
laterality	Which side	LEFT, RIGHT
severity	How severe	mild, moderate, severe

Procedure Event

One event per distinct procedure. Operative details like laser settings and access approach are captured as attributes rather than separate events.

Attribute	What It Records	Example Values
laterality	Which side	LEFT, RIGHT, BILATERAL
status	Completed or planned	completed, attempted, aborted
approach	Surgical approach	percutaneous, retrograde, antegrade, robotic
indication	Why it was done	decompression, stone removal, drainage
laser_power, laser_frequency	Laser parameters	0.8 J, 10 Hz

Device Event

Devices are tracked separately from procedures because "was a stent left in place?" and "what procedure was done?" are different research questions.

Attribute	What It Records	Example Values
device_type	Kind of device	ureteral stent, nephrostomy tube, Foley catheter
laterality	Which side	LEFT, RIGHT
status	Current state	placed, removed, in situ, exchanged
size	Device measurement	6 (paired with unit: Fr)

Medication Exposure

One event per distinct drug. Medications relevant to stone care include pain control, alpha-blockers for medical expulsive therapy, antibiotics, antiemetics, and stone prevention drugs.

Attribute	What It Records	Example Values
drug_class	Pharmacologic class	opioid, NSAID, alpha-blocker, antiemetic, antibiotic
dose	Amount given	0.4 mg, 15 mg, 500 mg
route	How administered	IV, PO, IM
frequency	How often	daily, BID, Q8H, PRN
context	Setting of administration	ED, intra-operative, at home, inpatient
indication	Why this drug was given	pain management, stone prevention, nausea/vomiting

Adverse Effect, Outcome Event, Recommendation Event

These three types cover the results of care. Adverse effects are complications (AKI, pyelonephritis, stent symptoms). Outcome events are treatment results (stone passage, stone-free status). Recommendation events are instructions for what comes next (follow-up, diet changes, strain urine).

Event Type	Key Attributes	Example
adverse_effect	reaction_type, severity, suspected_cause, timing, status	AKI, severity=moderate, suspected_cause=obstruction
outcome_event	status, value, qualifier, timing	stone-free status, value=true
recommendation_event	recommendation_type, target, status	follow_up, target=urology, status=scheduled

History Fact and Social Context Fact

History facts capture what happened before this visit: prior stones, prior surgeries, family history of stones, anatomic variants like horseshoe kidney. Social context facts capture support barriers: does the patient have transportation? Is a caregiver available? These are tracked because they affect treatment decisions and discharge planning.

Event Type	Key Attributes	Example
history_fact	history_kind, subject, count, intervention_status	kidney stone history, count=3, subject=PATIENT
social_context_fact	support_type, value	support_type=caregiver, value=available

4. Side-by-Side: Raw Text to Structured Events

These four examples show the system in action. On the left is the raw note text with key phrases highlighted. On the right is the structured output: the events extracted and the questions they answer. Each example demonstrates a different clinical scenario.

Example 1: ED Visit with Ureteral Stone

ED note. Acute stone, four medications, follow-up recommendation. Colors link raw text to the events they produce.

RAW NOTE TEXT

```
...male with past medical history
significant for kidney stones who presents
to the ED with acute right flank pain as
well as associated diffuse abdominal pain
with nausea and vomiting.

Punctate 2 mm nonobstructing calculus at the
right ureterovesicular junction.

ondansetron (ZOFRAN) injection 4 mg
intraVENOUS HYDROmorphone (DILAUDID)
injection 0.5 mg intraVENOUS ketorolac
(TORADOL) injection 15 mg intravenous

discharged with 5 mg oxycodone for pain
control

a follow-up with urology
```

STRUCTURED OUTPUT

EV1: Imaging Finding

Field	Value
preferred_term	ureteral stone
laterality	RIGHT
location	ureter (UVJ)
size / unit	2 mm
status	nonobstructing
stone_burden	punctate

EV2: Symptom Presentation

Field	Value
preferred_term	stone-related symptomatic presentation
symptoms	right flank pain, abdominal pain, nausea and vomiting
laterality	RIGHT

EV3-EV6: Medications

Drug	Dose	Route	Class
ondansetron	4 mg	IV	antiemetic
hydromorphone	0.5 mg	IV	opioid
ketorolac	15 mg	IV	NSAID
oxycodone	5 mg	PO	opioid

EV7: Recommendation

Field	Value
preferred_term	urology follow-up
type	follow_up

Questions Answered from This Note

Registry Question	Answer	Source
Was the patient symptomatic?	Yes	EV2
What were the symptoms?	right flank pain, abdominal pain, nausea and vomiting	EV2
Where is the stone?	ureter	EV1
What is the laterality?	RIGHT	EV1
What medication was given?	ondansetron, hydromorphone, ketorolac, oxycodone	EV3-6
What was the dose?	4 mg, 0.5 mg, 15 mg, 5 mg	EV3-6
What was the route?	IV, PO	EV3-6
What was the context?	ED	EV3-6

One note, 7 events, 3 families, 8 answered questions. Every answer traces back to an event, and every event traces back to the colored highlight in the raw text.

Example 2: Discharge Summary with Procedure and History

Discharge note. Current stone, cystoscopy, prior stone history, follow-up. Relations link the procedure to the stone it treated.

RAW NOTE TEXT

```
Nephrolithiasis [N20.0]

57-year-old woman with CAD, COPD, and
history of nephrolithiasis transferred from
an outside hospital for further management
of obstructive nephrolithiasis

Patient underwent cystoscopy with
decompression of the left side but will need
a repeat cystoscopy for stone retrieval.

Patient will follow up in urology clinic on
12/22/2020.
```

STRUCTURED OUTPUT

EV1: Imaging Finding

Field	Value
preferred_term	kidney stone
temporal_status	CURRENT

EV2: Procedure Event

Field	Value
preferred_term	cystoscopy
laterality	LEFT
indication	decompression
relation	performed_for -> EV1

EV3: Recommendation

Field	Value
preferred_term	urology follow-up
relation	follow_up_for -> EV1

EV4: History Fact

Field	Value
preferred_term	kidney stone history
history_kind	kidney stone history
temporal_status	HISTORICAL

Questions Answered from This Note

Registry Question	Answer	Source
Has the patient ever had a kidney stone before?	Yes	EV4

The cystoscopy (EV2) links to the stone (EV1) with a **performed_for** relation. The follow-up (EV3) links back to EV1 with **follow_up_for**. The history fact (EV4) is tagged HISTORICAL, so it answers "has the patient ever had a stone before?" but not "does the patient have a stone now?" Same concept, different temporal tag, different query result.

4. Side-by-Side (continued)

Example 3: Radiology Report. Confirmed vs. Negated Findings

Radiology report. Hydronephrosis is present. Ureteral stone is explicitly ruled out. Both extracted, different assertion strengths.

RAW NOTE TEXT

`Right renal hydronephrosis and ureteral dilation without definite finding of ureteral calculus.`

STRUCTURED OUTPUT

EV1: Imaging Finding

Field	Value
preferred_term	hydronephrosis
laterality	RIGHT
assertion	CONFIRMED

EV2: Imaging Finding

Field	Value
preferred_term	ureteral stone
assertion	NEGATED

Note. "Right renal hydronephrosis" is a positive finding. CONFIRMED. "Without definite finding of ureteral calculus" is explicit negation. NEGATED. Hydronephrosis without an identified stone is itself clinically meaningful. It could be a passed stone, a radiolucent stone, or a non-stone etiology. The projection system uses assertion_strength to route these correctly: CONFIRMED hydronephrosis answers "Is hydronephrosis present?" positively. The NEGATED stone does not answer "Where is the stone?"

Example 4: Inpatient Note with Social Context

Inpatient note. Post-procedure. Historical procedure, current device still in place, pain, caregiver documented.

RAW NOTE TEXT

```
...5 after percutaneous nephrostomy tube
placement...

Altered mental status, Pain...

Supine - in bed, Caregiver present...
```

STRUCTURED OUTPUT

EV1: Symptom Presentation

Field	Value
preferred_term	stone-related symptomatic presentation
symptoms	pain

EV2: Procedure Event

Field	Value
preferred_term	nephrostomy placement
approach	percutaneous
temporal_status	HISTORICAL

EV3: Device Event

Field	Value
preferred_term	nephrostomy tube
device_type	nephrostomy tube
temporal_status	CURRENT (still in place)

EV4: Social Context Fact

Field	Value
preferred_term	caregiver support
support_type	caregiver
value	available

Questions Answered from This Note

Registry Question	Answer	Source
Was the patient symptomatic?	Yes	EV1
What were the symptoms?	pain	EV1
What was used for drainage?	nephrostomy tube	EV3
Is there a caregiver at home?	Yes	EV4

Same source phrase produces two events. The procedure (EV2) is HISTORICAL because the placement already happened. The device (EV3) is CURRENT because the tube is still in the patient. Caregiver availability (EV4) becomes a queryable fact for discharge planning research.

5. From Events to Answers: The Registry

Events are the raw material. The registry is what makes them useful for research. 130 questions, designed by clinicians, imported from a prioritization workbook where each question was scored for clinical importance. The system computes answers automatically by looking at the events in each note.

What a Registry Question Looks Like

Each question has a unique ID, the original text as written by clinicians, the family it belongs to, and instructions for how to compute the answer from events. Here is the structure:

Field	What It Stores	Example
registry_id	Unique question identifier	what_medication_was_given
clinician_text	Original question as written	"What medication was given?"
family	Which clinical domain answers it	medications
answer_scope	Per-note or across all notes	note (or patient_timeline)
answer_shape	Type of answer returned	boolean, list, count, or grouped_count
resolver_id	Logic for computing the answer	event_concepts, event_presence, etc.

Answer Shapes

There are four ways a question can be answered. The shape determines what the answer looks like and how the system computes it from the extracted events:

Shape	What It Returns	Example Question	Example Answer
boolean	Yes or No	"Was the patient symptomatic?"	true
list	One or more values	"What medication was given?"	["tamsulosin", "morphine"]
count	A number	"How many stones passed?"	3
grouped_count	Values grouped with counts	"Medications by class"	[{"opioid": 2}, {"NSAID": 1}]

How the Answer Is Stored: the Projection

When the system computes an answer, it stores it as a "projection." Think of it as the receipt: it records which question was answered, what the answer is, and which events prove it.

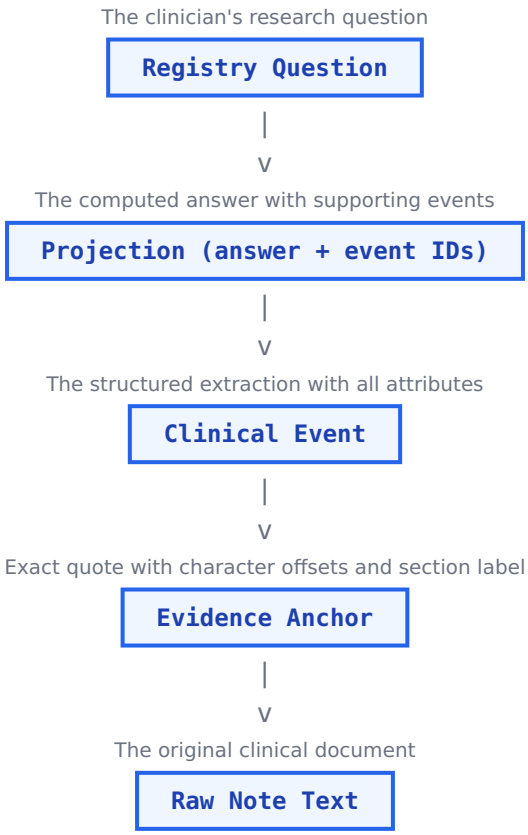
Field	What It Stores	Example
registry_id	Which question this answers	what_medication_was_given
scope_key	Which note (or patient timeline)	ED.IDX.0000132
resolution_state	Was it answerable?	answered (or unanswered)
value	The answer itself	["hydromorphone", "ketorolac", "ondansetron", "oxycodone"]
supporting_event_ids	Which events prove this answer	[EV3, EV4, EV5, EV6]

A Real Projection

Here is what the system actually stores when it answers "What medication was given?" for the ED visit from Example 1:

```
registry_id:           what_medication_was_given
scope_key:             ED.IDX.0000132
resolution_state:      answered
value:                 ["hydromorphone", "ketorolac", "ondansetron", "oxycodone"]
supporting_event_ids:  [EV3, EV4, EV5, EV6]
```

A researcher can see the answer (four drugs) and immediately look up EV3 through EV6 to see exactly where in the note each drug was mentioned. This is the full provenance chain:


The clinician's research question
Registry Question
The computed answer with supporting events
Projection (answer + event IDs)
The structured extraction with all attributes
Clinical Event
Exact quote with character offsets and section label
Evidence Anchor
The original clinical document
Raw Note Text

6. Building Cohort Definitions

Because every note produces structured events and answered questions, you can query across all 764 annotated notes to define patient cohorts for research. A cohort definition is just a filter on the structured data.

Example Queries

Patients with obstructing ureteral stones who received tamsulosin

```
family = symptoms_imaging, event_type = imaging_finding, preferred_term = "ureteral stone", attributes.status = "obstructing"
```

AND

```
family = medications, event_type = medication_exposure, preferred_term = "tamsulosin"
```

JOIN on patient_id

This query works because the stone's status (obstructing vs. nonobstructing) is stored as a structured attribute on the imaging_finding event, and each medication is its own event with a canonical preferred_term. You join across families on patient_id.

Patients who had ureteroscopy and developed a post-operative complication

```
family = procedure_devices, event_type = procedure_event, preferred_term contains "ureteroscopy"
```

AND

```
family = outcomes_complications, event_type = adverse_effect
```

JOIN on patient_id

Stone-free rate after ureteroscopy

```
registry_id = "stone_free_status", resolution_state = "answered"
```

FOR patients with

```
event_type = procedure_event, preferred_term contains "ureteroscopy"
```

The third query demonstrates why the projection layer matters. Instead of scanning all events to infer "stone-free," you query the pre-computed answer to the "stone-free status" question directly. The projection already did the work of checking whether the right outcome_event exists for each note.

When the structured events are flattened into tabular form for analysis, each row is one note with its key clinical attributes. This is what a researcher exports to a spreadsheet or statistical tool:

patient_id	note_date	stone_location	laterality	size	procedure	symptomatic	medications	complications
HID.972...	2025-02-13	ureter (UVJ)	RIGHT	2 mm	-	Yes	ondansetron, hydromorphone, ketorolac, oxycodone	-
HID.112...	2022-12-31	distal ureter	RIGHT	5 mm	-	Yes	morphine, tamsulosin, ketorolac	-
HID.108...	2020-12-18	kidney	-	-	cystoscopy (LEFT)	-	-	-
HID.907...	2018-04-03	kidney	-	-	stent + nephrostomy	-	-	acute kidney injury

7. Connecting Events: Relations and Context Labels

Relations Between Events

Events within the same note can be linked to each other. A procedure can be linked to the stone it was treating. A complication can be linked to the procedure that caused it. A follow-up recommendation can be linked to the condition it is monitoring. These links let researchers ask questions like "what complications occurred after ureteroscopy?" by following the relation edges.

Relation Type	Meaning	Example
performed_for	Procedure done to treat a finding	ureteroscopy performed_for ureteral stone
caused_by	Complication caused by a treatment	AKI caused_by obstruction
follow_up_for	Recommendation monitors a condition	urology follow-up follow_up_for kidney stone
treated_by	Finding treated by a procedure	hydronephrosis treated_by stent placement
decompresses	Device relieves obstruction	nephrostomy tube decompresses hydronephrosis
temporally_linked	Events happened in sequence	stent placed temporally_linked to ureteroscopy
modifies	One event refines another	stone analysis modifies imaging finding
resolved_after	Condition resolved after treatment	hydronephrosis resolved_after stent placement
causes_obstruction	Stone causes urinary obstruction	ureteral stone causes_obstruction

Each relation records which two events are connected, the type of connection, and a brief rationale explaining why the link exists. Relations are stored as part of the source event and point to the target event by ID.

Temporal Status

Every event is tagged with when it happened relative to the current visit. This is critical for answering time-sensitive questions correctly:

Value	Meaning	Note Text Example
CURRENT	This visit or actively present now	"5mm stone in right ureter"
HISTORICAL	Happened before this visit	"history of nephrolithiasis"
FUTURE	Planned but not yet done	"will undergo ureteroscopy next week"

> A HISTORICAL procedure does not answer "what procedure was performed this visit?" A FUTURE recommendation is not a completed one. Without temporal tagging, a query for "patients who had ureteroscopy" would incorrectly include patients who only had it planned or who had it years ago.

Assertion Strength

Clinicians do not always state things with certainty. The system captures three levels:

Value	Meaning	Note Text Example
CONFIRMED	Definitively present or performed	"CT shows 5mm calculus"
POSSIBLE	Uncertain, pending confirmation	"possible robotic pyeloplasty pending results"
NEGATED	Explicitly ruled out by the clinician	"without definite finding of ureteral calculus"

NEGATED events are extracted and stored, but they do not produce positive answers to clinical questions. This preserves the clinically important distinction between "we looked and it was not there" and "we did not look."

Experiencer

Most events happen to the patient, but some notes mention family history of kidney stones. The experiencer field tracks who:

Value	When Used
PATIENT	The event happened to the patient (default, vast majority of events)
FAMILY_MEMBER	Family history: "mother has kidney stones"

Projection Exclusion Rules

Not every extracted event contributes a positive answer to a question. The projection system applies exclusion rules so that only clinically appropriate events produce results:

Condition	Effect on Projection
NEGATED assertion	Excluded from positive answers ("stone not seen" does not answer "Where is the stone?")
FUTURE temporal status	Excluded. Planned events have not occurred yet.
HISTORICAL temporal status	Excluded from *note-scope* questions (but included in timeline-scope)
PROVISIONAL / CUSTOM events	Excluded. Only fully validated events project.
Procedure status: planned / scheduled / deferred / canceled	Excluded from procedure-completion questions

> Without exclusion rules, a query for "was ureteroscopy performed?" would count future planned procedures and negated findings as positive answers. The rules ensure projections reflect what actually happened to the patient during the encounter.

Artifact Shape: What Gets Stored

Each annotated note produces a JSON artifact containing the full extraction:

```
{
  "schema_version": "event-first",
  "patient_id": "HID.907...",
  "note_id": "OUT.IDX.0004721",
  "events": [
    {
      "event_id": "evt-...",
      "event_type": "imaging_finding",
      "preferred_term": "hydronephrosis",
      "assertion_strength": "CONFIRMED",
      "temporal_status": "CURRENT",
      "attributes": { "laterality": "RIGHT", "severity": "moderate" },
      "evidence": { "anchors": [{ "quote": "Right renal hydronephrosis", ... }] },
      "field_provenance": {
        "preferred_term": { "source": "model", "overwrote_model": false },
        "assertion_strength": { "source": "model", "overwrote_model": false },
        "laterality": { "source": "candidate_binding", "overwrote_model": false }
      },
      "relations": [...]
    }
  ],
  "projections": [
    { "registry_id": "Q041", "answer": ["RIGHT"], "resolver": "event_attribute_values" }
```

```
    ]
  }
```

The **field_provenance** block records how each field was set. Whether the model proposed it, deterministic enrichment filled it, or a guardrail overwrote the model. This audit trail enables quality analysis: if a guardrail frequently overrides the model on a field, the extraction prompt needs refinement.

8. The 130 Questions by Domain

The registry questions were designed by clinicians and prioritized through a voting process. Each question maps to exactly one of the five families. The distribution reflects clinical emphasis:

Family	Questions	Why
Procedures & Devices	**66**	Largest. Surgical detail drives most treatment-level research.
History & Timeline	**31**	Prior stone history, comorbidities, social factors
Outcomes & Complications	**17**	Complication rates, stone-free status, follow-up
Symptoms & Imaging	**9**	Stone characterization (size, location, obstruction)
Medications	**7**	Drug class, dose, route, indication
Total	**130**	

The complete list follows, grouped by family. Each question shows the resolver that computes the answer and the expected answer shape. Duplicate entries appear when the clinician workbook listed the same question under multiple procedure sub-contexts (PCNL vs. URS vs. SWL).

Procedure & Devices (66 questions)

#	Question	Resolver	Shape
1	Was genetic testing for kidney stones performed?	`event_presence`	categorical
2	Was a post-operative stent left?	`event_presence`	categorical
3	Was FANS used?	`event_presence`	categorical
4	How many tracts were used?	`event_attribute_values`	numeric
5	What was the PCNL tract size?	`event_attribute_values`	numeric
6	What was the shock rate used?	`event_attribute_values`	numeric
7	Which service obtained percutaneous access?	`event_attribute_values`	categorical
8	What was the genetic testing result?	`event_attribute_values`	categorical
9	Was a ureteral access sheath used?	`event_presence`	categorical
10	Was the URS aborted due to inability to access the ureter?	`event_presence`	categorical
11	What was used for drainage at the end of the case?	`event_concepts`	categorical
12	Was the stent left on a string/tether?	`event_attribute_values`	categorical
13	What was the type of laser used?	`event_attribute_values`	categorical
14	Was suction used?	`event_presence`	categorical
15	Was CVAC used?	`event_presence`	categorical
16	Was ECIRS used?	`event_presence`	categorical
17	What was the diameter of the UAS?	`event_attribute_values`	numeric
18	Was balloon dilation used?	`event_presence`	categorical
19	What was the maximum energy?	`event_attribute_values`	categorical
20	What was the total energy?	`event_attribute_values`	categorical
21	What was the type of stent used?	`event_attribute_values`	categorical
22	What was the length of the stent?	`event_attribute_values`	numeric
23	What was the laser setting power?	`event_attribute_values`	numeric
24	How was the patient positioned?	`event_attribute_values`	categorical
25	Was DISS used?	`event_presence`	categorical

#	Question	Resolver	Shape
26	Was a ramp up energy strategy used?	`event_presence`	numeric
27	What was the ureteroscope type?	`event_attribute_values`	categorical
28	What was the diameter of the stent?	`event_attribute_values`	numeric
29	What was the diameter of the balloon dilation?	`event_attribute_values`	numeric
30	What was the laser fiber type?	`event_attribute_values`	categorical
31	What was the make / model of ureteroscope?	`event_attribute_values`	categorical
32	How was the patient positioned?	`event_attribute_values`	categorical
33	Was DISS used?	`event_presence`	categorical
34	Was a 2 minute pause used?	`event_presence`	categorical
35	What was the laser setting power?	`event_attribute_values`	numeric
36	Was a ramp up energy strategy used?	`event_presence`	numeric
37	What was the lithotripsy technique?	`event_concepts`	categorical
38	What was the type of SWL machine?	`event_attribute_values`	categorical
39	What was the irrigation method?	`event_attribute_values`	categorical
40	What was the ureteroscope type?	`event_attribute_values`	categorical
41	What was the diameter of the stent?	`event_attribute_values`	numeric
42	What was the laser setting frequency?	`event_attribute_values`	numeric
43	Was a retrograde pyelogram performed?	`event_presence`	categorical
44	What was the size of the ureteroscope?	`event_attribute_values`	numeric
45	What were the findings of the retrograde pyelogram?	`event_attribute_values`	list
46	What was the laser fiber size?	`event_attribute_values`	numeric
47	Was coaxial dilation used?	`event_presence`	categorical
48	What was the length of the stent?	`event_attribute_values`	numeric
49	What was the diameter of the balloon dilation?	`event_attribute_values`	numeric
50	What was the lithotripter energy source?	`event_attribute_values`	categorical

#	Question	Resolver	Shape
51	What was the irrigation pressure?	`event_attribute_values`	numeric
52	Was a retroperc used?	`event_presence`	categorical
53	What was the laser setting frequency?	`event_attribute_values`	numeric
54	What was the laser setting power?	`event_attribute_values`	numeric
55	What was the diameter of coaxial dilation?	`event_attribute_values`	numeric
56	What was the length of the UAS?	`event_attribute_values`	numeric
57	Was a safety wire used?	`event_presence`	categorical
58	What was the laser fiber type?	`event_attribute_values`	categorical
59	What was the laser fiber type?	`event_attribute_values`	categorical
60	What was the suction pressure?	`event_attribute_values`	numeric
61	What was the make / model of ureteroscope?	`event_attribute_values`	categorical
62	What was the laser fiber size?	`event_attribute_values`	numeric
63	Was a dual lumen catheter used	`event_presence`	categorical
64	What was the diameter of the dual lumen?	`event_attribute_values`	numeric
65	What irrigation type was used?	`event_attribute_values`	categorical
66	Was a safety wire used?	`event_presence`	categorical

History & Timeline (31 questions)

#	Question	Resolver	Shape
1	Has the patient ever had a kidney stone before index visit?	`event_presence`	timeline_fact
2	How many stones has the patient had prior to the index visit?	`event_attribute_values`	numeric
3	Has the patient ever had kidney stone surgery prior to the index visit?	`event_presence`	timeline_fact
4	What was the age of onset of first stone prior to the index visit?	`event_attribute_values`	numeric
5	Has the patient ever passed a stone prior to the index visit?	`event_presence`	timeline_fact
6	What urologic surgeries have the patients received prior to the index visit?	`event_concepts`	list
7	Has the patient had recurrent UTIs?	`event_presence`	categorical
8	Is there a family history of kidney stones?	`event_presence`	timeline_fact
9	What was the prior stone surgery type prior to the index visit?	`event_concepts`	timeline_fact
10	How many kidney stones has the patient passed?	`event_attribute_values`	numeric
11	Does the patient have transportation?	`event_presence`	categorical
12	What anatomic variants does the patient have?	`event_concepts`	categorical
13	Has the patient had a prior UTI?	`event_presence`	timeline_fact
14	Has the patient ever had a ureteral stent?	`event_presence`	timeline_fact
15	Has the patient had MRDO UTIs?	`event_presence`	categorical
16	How many prior stone surgeries were performed (of each type)	`event_grouped_counts`	numeric
17	Has the patient had kidney stones during pregnancy?	`event_presence`	categorical
18	What is the patient's frailty index?	`event_attribute_values`	categorical
19	Does the patient have food insecurity?	`event_presence`	categorical
20		`event_attribute_values`	list

#	Question	Resolver	Shape
	Which family members have a history of kidney stones?		
21	Has the patient been pregnant?	`event_presence`	categorical
22	What is the profession of the patient?	`event_attribute_values`	categorical
23	Is the patient on enteral nutrition?	`event_presence`	categorical
24	What is the education level of the patient?	`event_attribute_values`	categorical
25	Which family members have a history of kidney stones?	`event_attribute_values`	list
26	Is the patient on enteral nutrition?	`event_presence`	categorical
27	What type of tube feeds are used?	`event_presence`	categorical
28	Has the patient been pregnant?	`event_presence`	categorical
29	What is the education level of the patient?	`event_attribute_values`	categorical
30	What is the profession of the patient?	`event_attribute_values`	categorical
31	Is there a caregiver that helps at home?	`event_presence`	categorical

Outcomes & Complications (17 questions)

#	Question	Resolver	Shape
1	What medical side effects were encountered?	`event_concepts`	categorical
2	Was there an intra-operative ureteral injury?	`event_presence`	categorical
3	What was the recommended diet change?	`event_concepts`	categorical
4	Was post-operative obstruction present?	`event_presence`	categorical
5	What was the baseline diet?	`event_attribute_values`	categorical
6	Did the patient experience urinary retention?	`event_presence`	categorical
7	Was a hematoma present after surgery	`event_presence`	categorical
8	Was a urine leak present after surgery?	`event_presence`	categorical
9	Did the patient experience an anesthetic complication?	`event_concepts`	categorical
10	Did the patient experience urinary retention?	`event_presence`	categorical
11	Was a hematoma present after surgery	`event_presence`	categorical
12	Did the patient experience an obstetric complication?	`event_presence`	categorical
13	Has the patient had any complications during pregnancy?	`event_presence`	categorical
14	Was increased physical activity recommended?	`event_presence`	categorical
15	Was the patient stone free or were there residual fragments after treatment?	`event_concepts`	categorical
16	Did the patient experience stone passage?	`event_presence`	categorical
17	Did the patient's symptoms resolve after treatment?	`event_presence`	categorical

Symptoms & Imaging (9 questions)

#	Question	Resolver	Shape
1	Is the stone a staghorn calculus?	`event_presence`	categorical
2	What is the Hounsfield unit for the stone?	`event_attribute_values`	numeric
3	Was the patient symptomatic from this stone	`event_presence`	categorical
4	Where is the stone (stone location)?	`event_attribute_values`	categorical
5	Is hydronephrosis present?	`event_presence`	categorical
6	What were the patient's symptoms?	`event_attribute_values`	list
7	What is the laterality of the stone?	`event_attribute_values`	categorical
8	Was hydronephrosis present?	`event_presence`	categorical
9	What imaging modality was used to guide access?	`event_attribute_values`	categorical

Medications (7 questions)

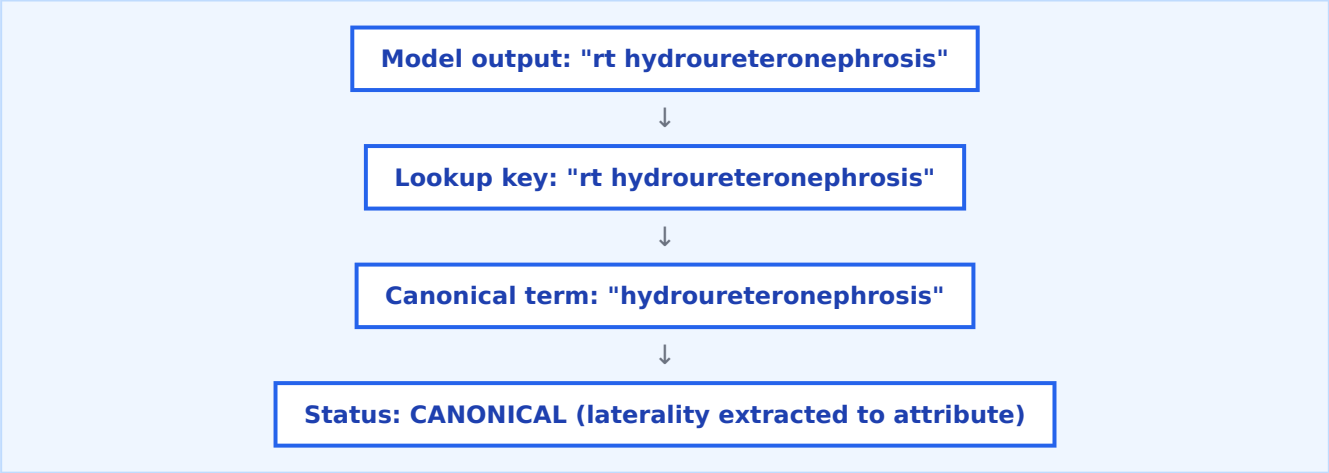
#	Question	Resolver	Shape
1	What medication was given?	`event_concepts`	list
2	What was the date of administration / prescription?	`event_attribute_values`	categorical
3	What was the duration of the medication?	`event_attribute_values`	categorical
4	What was the frequency of dosing (i.e. once a day; twice a day)	`event_attribute_values`	numeric
5	What was the dose of medication	`event_attribute_values`	categorical
6	What was the context of medication administration (i.e. at home, intra-operatively, pre-op)	`event_attribute_values`	categorical
7	What was the route of medication administration	`event_attribute_values`	categorical

9. Controlled Vocabularies

Raw clinical text is inconsistent. A surgeon writes "URS," another writes "ureteroscopy," a third writes "flexible ureteroscopy." All three mean the same procedure. The system handles this through two parallel vocabulary layers: one for event preferred terms, one for attribute values.

How Normalization Works

When the extraction model proposes an event, the normalization engine receives the raw preferred_term and each attribute value. It lowercases and strips punctuation to build a lookup key, then checks that key against the vocabulary tables. If a match exists, the raw text is replaced with the canonical form.



If no alias matches, the event gets a PROVISIONAL status and a generated concept_id. PROVISIONAL events are valid extractions. They just indicate the vocabulary has not seen this exact phrasing before. The gap-reporting system flags these so new aliases can be added over time.

Event Preferred Terms

The event vocabulary maps 990 aliases to 252 canonical preferred terms across all 10 event types. Each (family, event_type) pair has its own lookup table. A few examples from the imaging_finding vocabulary:

```
"ureteral calculus" → "ureteral stone" "uvj stone" → "ureteral stone" "distal ureteral stone" →
"ureteral stone" "obstructing ureteral calculus" → "ureteral stone" "mild right hydronephrosis" →
"hydronephrosis" "moderate left hydronephrosis" → "hydronephrosis" "staghorn calculi" → "staghorn
calculus"
```

The canonical term ("ureteral stone") is what gets stored in the event. Laterality, severity, and location are parsed into separate attributes. This separation prevents a combinatorial explosion. Instead of maintaining entries for every combination of laterality + severity + finding, the system normalizes the finding once and captures the modifiers as structured attribute values.

Attribute Vocabularies

The attribute vocabulary covers 57 controlled attributes across all event types, mapping 910 aliases to 327 canonical values. Each attribute is scoped to a specific (family, event_type, attribute) triple. Here is how the symptom vocabulary works:

```
"left flank pain" → "LEFT FLANK PAIN" (upper_tokens style) "l flank pain" → "LEFT FLANK PAIN" "cva
tenderness" → "CVA TENDERNESS" "costovertebral angle tenderness" → "CVA TENDERNESS" "gross
hematuria" → "HEMATURIA" "blood in urine" → "HEMATURIA" "colicky flank pain" → "RENAL COLIC"
```

Canonical style varies by attribute. Symptoms use UPPER_TOKENS because they appear in structured output tables alongside coded values. Medication routes use the same convention (IV, PO, IM). Drug classes use lower_text (opioid, NSAID, alpha-blocker) because they read more naturally in prose summaries.

Vocabulary Coverage by Domain

The table below shows the largest attribute vocabularies. These are the attributes that see the most variation in clinical text and require the most aliases to normalize consistently.

Family	Event Type	Attribute	Canonical Values	Aliases
symptoms_imaging	symptom_presentation	symptoms	34	76
history_timeline	history_fact	history_kind	12	56
outcomes_complications	adverse_effect	reaction_type	26	52
medications	medication_exposure	frequency	15	51
medications	medication_exposure	drug_class	23	45
symptoms_imaging	imaging_finding	location	17	42
procedure_devices	device_event	device_type	18	34
outcomes_complications	recommendation_event	recommendation_type	9	30
symptoms_imaging	imaging_finding	comparison_status	8	24
medications	medication_exposure	indication	7	25

Vocab Guidelines: The Attribute Policy Layer

Each of the 57 attributes has an explicit policy that defines three things: what kind of vocabulary it uses, how canonical values are formatted, and whether the extraction model is allowed to introduce new values.

Property	Options	What It Controls
attribute_kind	controlled, semi_structured, free_text	Whether values must come from the vocabulary (controlled), follow a pattern like dates or measurements (semi_structured), or allow any text (free_text)
canonical_style	upper_tokens, lower_text, lower_or_snake, clinical_mixed, measurement_unit, numeric_or_text, date_or_span, free_text	The formatting convention for canonical values
allow_provisional	true / false	Whether unrecognized values are accepted (true) or rejected (false). Controlled attributes with allow_provisional=false will reject any value not in the vocabulary.

A concrete example. The attribute `imaging_finding.laterality` has kind=controlled, style=upper_tokens, allow_provisional=false. That means the only legal values are LEFT, RIGHT, and BILATERAL. If the model produces "rt" or "bilateral kidneys," the normalization engine maps it to the canonical form. If it produces something unmappable, the value is rejected and flagged.

Contrast that with `procedure_event.indication`: kind=controlled, style=lower_text, allow_provisional=true. The vocabulary has 9 canonical values (stone removal, stent placement, drainage, etc.), but if the model encounters a genuinely new indication like "stent exchange," it can accept the provisional value while the gap reporter flags it for vocabulary expansion.

> The vocabulary system trades upfront curation effort for downstream consistency. When a researcher queries "all patients with hydronephrosis," they get every instance. Not just the ones where the surgeon happened to write "hydronephrosis" rather than "mild right hydroureteronephrosis." The 910 attribute aliases and 990 event term aliases are what make that guarantee possible.